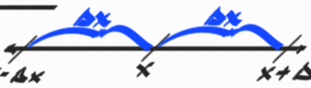
Boundary value Problem: Finite Difference



Finite Difference uses the combination of Forward + Backward Difference methods.

F.D) $f(x+\Delta x) = f(x) + \Delta x f'(x) + \frac{\Delta x^2 f''(x)}{2!} + \frac{\Delta x^3 f'''(x)}{3!} + \dots + \frac{\Delta x^n f^n(x)}{n!}$

B.D) $f(x-\Delta x) = f(x) - \Delta x f'(x) + \frac{\Delta x^2 f''(x)}{2!} - \frac{\Delta x^3 f'''(x)}{3!} + \dots \pm \frac{\Delta x^n f^n(x)}{n!}$

Finite Difference Process

*** Solve for $f'(x)$**

$$f(x+\Delta x) - f(x-\Delta x) = 2\Delta x f'(x) + \frac{2\Delta x^3 f'''(x)}{3!}$$

$$\Rightarrow f'(x) = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x} - \frac{\Delta x^2 f'''(x)}{3!}$$

*** Solve for $f''(x)$**

$$f(x+\Delta x) + f(x-\Delta x) = 2f(x) + \Delta x^2 f''(x) + \frac{2\Delta x^4 f''''(x)}{4!}$$

$$\Rightarrow f''(x) = \frac{f(x+\Delta x) - 2f(x) + f(x-\Delta x)}{\Delta x^2} - \frac{2\Delta x^2 f''''(x)}{4!}$$

- Approximation
- Remainder $\Rightarrow \Theta(\Delta x^2)$
- $\Rightarrow$ 2nd order accuracy.

* Note: $\Theta(\Delta x^2) \Rightarrow$ Small angle Approximation can be neglected as it will be significantly small enough to negligible

$\therefore$ Finite General Form Approximation: $f'(x) = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x}$

$$f''(x) = \frac{f(x+\Delta x) - 2f(x) + f(x-\Delta x)}{\Delta x^2}$$

* Note: to Descritize

- $f'(x) \Rightarrow f'(x_i) \Rightarrow y_i'$
- $f''(x) \Rightarrow f''(x_i) = y_i''$
- $f(x+\Delta x) \Rightarrow f(x_i+\Delta x) = y_{i+1}$
- $f(x-\Delta x) \Rightarrow f(x_i-\Delta x) = y_{i-1}$

Example: Linear ODE: $y'' + xy' - xy = 2x$ w/ BC $y(0)=1$ $y(1)=8$ $\Delta x = 0.5$

Apply Finite Difference

$$\Rightarrow \frac{y_{i+1} - 2y_i + y_{i-1}}{\Delta x^2} + x_i\left(\frac{y_{i+1} - y_{i-1}}{2\Delta x}\right) - x_i y_i = 2x_i$$

$$\Rightarrow y_{i+1} - 2y_i + y_{i-1} + \frac{x_i \Delta x\,(y_{i+1} - y_{i-1})}{2} - x_i \Delta x^2 y_i = 2x_i \Delta x^2$$

$$\Rightarrow y_{i+1}\left(1 + x_i\frac{\Delta x}{2}\right) - y_i\left(2 + x_i \Delta x^2\right) + y_{i-1}\left(1 - x_i\frac{\Delta x}{2}\right) = 2x_i\Delta x^2$$

* Note: $y_{i+1} - y_i + y_{i-1}$ = Non y's $\Rightarrow$ Recursion form for finite Difference

$$y_{\text{next}}: \begin{bmatrix} y_{RBC} & 0 & 0 & 0 & 0 \\ y_{2-1} & y_2 & y_{2+1} & 0 & 0 \\ 0 & y_{3-1} \dots & y_3 & y_{3+1} \dots & 0 \\ 0 & 0 \dots & y_{i-1} & y_i \dots & y_{i+1} \\ 0 & 0 & 0 & 0 & y_{RBC} \end{bmatrix} = \begin{bmatrix} Sol_1 \\ Sol_2 \\ Sol_3 \\ Sol_4 \\ Sol_n \end{bmatrix} \text{Non-}y\text{'s}$$

Note: Sol_1 = initial Cond. Sol_2 = final Cond.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ (1-x_i\frac{\Delta x}{2}) & (2+x_i\Delta x^2) & (1+x_i\frac{\Delta x}{2}) & 0 & 0 \\ 0 & (1-x_i\frac{\Delta x}{2}) & (2+x_i\Delta x^2) & (1+x_i\frac{\Delta x}{2}) & 0 \\ 0 & 0 & (1-x_i\frac{\Delta x}{2}) & (2+x_i\Delta x^2) & (1+x_i\frac{\Delta x}{2}) \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_{i+1} \\ y_{i+2} \\ y_{i+3} \\ y_{i+n} \\ y_{i+n+1} \end{bmatrix} = \begin{bmatrix} Sol_1 \\ Sol_2 \\ Sol_3 \\ Sol_4 \\ Sol_n \end{bmatrix}$$

$$[A][\vec{y}] = [\vec{b}]$$

Non-linear ODE

$$y'' = 3y + x^2 + 100y^2 \quad \text{B.C } y(0)=0,\ y(1)=0$$

Step 1: Apply finite difference

Step 2: Use iteration method to solve

* Solve the ODE w/o Non-linear term using the initial condition solution for the guess of y

* Then incorporate the guess sol in the Non-linear term on the right hand side treating it as a constant.

- * called y_{old}
- * Solve full equation for a y_{new}

* Check whether y_{new} matches y_{new} w/ some tolerance (10^{-6})

* $y_{old} \approx y_{new}$

PDE's: Partial Derivative equations, derivatives multi-Dimension

ex: $T(x, y, z, t)$

general 2^{nd} Order PDE:

$$u(x,y) \Big| A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x \partial y} + C\frac{\partial^2 u}{\partial y^2} = F\left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right)$$

*Note: Constants A, B, C Determine type of PDE

$B^2 - AC$ | < 0 Elliptic PDE
| $= 0$ Parabolic PDE
| > 0 Hyperbolic PDE

Elliptic PDE's

* Laplace Equation 2-D: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x^2} = \nabla^2 u = 0$ | $\nabla = \frac{\partial}{\partial x} + \frac{\partial}{\partial y}$ = gradient

* Heat Conduction in 2D: unsteady | $\frac{\partial u}{\partial t} = \alpha \nabla^2 u$ | α = thermal diffusivity, u = Temperature

* Poisson's equation: $\nabla^2 u = f(x,y)$ | $\frac{\partial u}{\partial t} = \alpha\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$

Methods: To solve Numerically, we can use finite Difference both x & y and discretize into grid.

Types of Boundary Conditions:

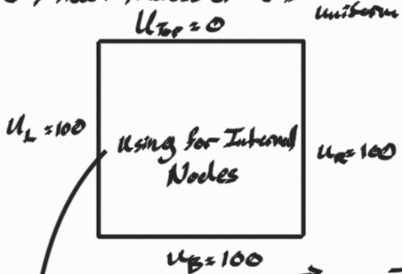
* Dirichlet: $y(0) = a$ $y(L) = b$ Specifying values @ Boundaries
* Neumann: $y'(0) = a$ $y'(L) = b$ Specifying derivatives @ Boundaries
* Robin: Specifies linear Combination of the function and its derivative

  $a\,y(0) + b\,y'(0) = g(0)$

  $a\,y(L) + b\,y'(L) = g(L)$
* Mixed: Specifies any combination of the above: $y(0) = 0$, $y'(0) = b$; $y(L) = a$, $y'(L) = b$

Dirichlet Boundary Condition

ex: Heat transfer 2-D s.s. uniform grid



$$\underbrace{\begin{bmatrix} 1&0&0&0&0&0&0&0&0 \\ 0&1&0&0&0&0&0&0&0 \\ 0&0&1&0&0&0&0&0&0 \\ 0&0&0&1&0&0&0&0&0 \\ 0&1&0&1&-4&1&0&0&0 \\ 0&0&0&0&0&1&0&0&0 \\ 0&0&0&0&0&0&1&0&0 \\ 0&0&0&0&0&0&0&1&0 \\ 0&0&0&0&0&0&0&0&1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} u_{1,1} \\ u_{2,1} \\ u_{3,1} \\ u_{1,2} \\ u_{2,2} \\ u_{3,3} \\ u_{1,3} \\ u_{2,3} \\ u_{3,3} \end{bmatrix}}_{\vec{u}} = \underbrace{\begin{bmatrix} 100 \\ 100 \\ 100 \\ 100 \\ 0 \\ 100 \\ 100 \\ 0 \\ 100 \end{bmatrix}}_{\vec{b}}$$

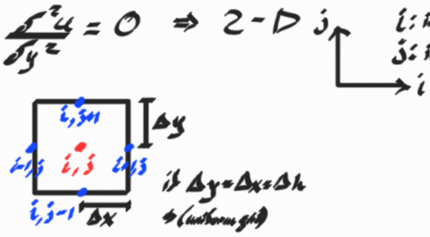
$$\vec{u} = \vec{b} \cdot \vec{A}^{-1}$$

ex) $\nabla^2 u = 0 \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow$ 2-D

i: indexes x
j: indexes y

$$\frac{\partial^2 u}{\partial x^2} = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{\Delta x^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{\Delta y^2}$$



if $\Delta y = \Delta x = \Delta h$ (uniform grid)

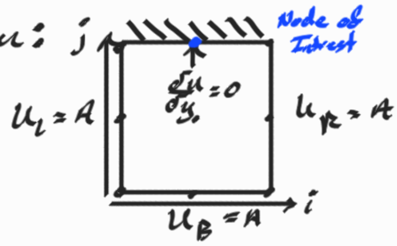
$$\Rightarrow \text{discritized form: } 0 = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{\Delta x^2} + \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{\Delta x^2}$$

$$\Rightarrow 0 = u_{i+1,j} + u_{i,j+1} + u_{i-1,j} + u_{i,j-1} - 4u_{i,j}$$

Row – Major Mapping : Defines a larger grid

$K_{i,j} = jn + i$ | K = 1-D index K mapped from 2-D indices i, j n = index of Nodes 1-D

Neuman Boundary Condition :



: for $\frac{\partial u}{\partial y}$ use Forward, Backward or Central Difference

ex: using Central Difference

$$\Rightarrow \frac{\partial u}{\partial y} = \frac{u_{i,j+1} - u_{i,j-1}}{2\Delta y} = 0$$

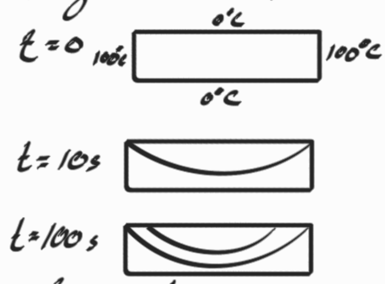
$$\Rightarrow u_{i,j+1} = u_{i,j-1}$$

∴ U vals in top row of mat = U vals 2 rows below top

Parabolic PDE's : Using I.V.P. in time, BVP in space

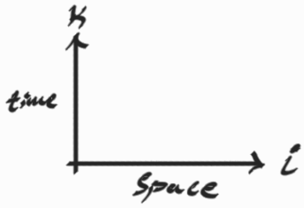
ex) 1-D unsteady heat equ.

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$



$$\Rightarrow \frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

Solve explicitly w/ central Difference in Space + forward Difference in time



to discretize: $\frac{T_{i-1}^{K} - 2T_i^{K} + T_{i+1}^{K}}{\Delta x^2} = \frac{1}{\alpha}\left(\frac{T_i^{K+1} - T_i^{K}}{\Delta t}\right)$

⇒ Combing like terms and Solve for T_i^{K+1} to get recursion form

* Note : for Parabolic PDE's

- * Explicit method: solve explicitly — F-D in time, C-D in space
  - F-D is conditionaly stable
  - Order of accuracy less then 2
- * Crank - Nicolson Method:
  - * Still use C-D in space but use finite Difference in time
    - order of accuracy will increase
    - Unconditionally stable

$$\Rightarrow \frac{u_i^{K+1} - u_i^{K}}{\Delta t} = F_i^{K} \Big|_{\text{for forward Diff.}} \qquad \frac{u_i^{K+1} - u_i^{K}}{\Delta t} = F_i^{K+1} \Big|_{\text{for Backward Diff.}}$$

∴ Crank - Nicolson : $\frac{u_i^{K+1} - u_i^{K}}{\Delta t} = \frac{F_i^{K} + F_i^{K+1}}{2}$